Computer Graphics Project Requirements

General Information

- The project will be completed in groups of **8 students**.
- Students from different sections are allowed to form one group.
- Total project grade: **30 marks**.
- Each group must submit:
    - A working project.
    - A presentation.
    - A documentation report.

Technical Requirements

All projects must be fully 3D-based. 2D projects will not be accepted.

Option 1: OpenGL (C++ only)

- The project must be implemented using **C++ with OpenGL** only.
- No bonus marks will be awarded for this option.
- Maximum grade: **30 marks**.

Option 2: Unity (C#) or Any Framework

- The project may be implemented using Unity (C#) or any other graphics framework (Unreal Engine).
- In all cases, the project must include clear programming implementation (scripting using C++, C#, or visual scripting such as Blueprints).
- Projects that rely only on pre-built assets or editor configurations without significant coding/scripting effort will not receive full marks.

A bonus of up to 3 extra marks may be awarded based on technical complexity, programming depth, and creativity

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Project Evaluation Criteria (30 Marks)

A) Implementation (24 Marks)

1. Correct use of transformations (translation, rotation, scaling).
2. Proper camera setup (orthographic or perspective).
3. Lighting and material implementation.
4. Texture mapping.
5. Object modeling quality.
6. Animation (if applicable).
7. Interaction (keyboard/mouse control).
8. Collision detection (if applicable).
9. Code organization and structure.
10. Overall technical complexity.

B) Documentation (3 Marks)

1. Cover page (project title, team members, IDs).
2. Project description.
3. Objectives.
4. System design.
5. Used technologies.
6. Explanation of implemented features.
7. Screenshots.
8. Code structure explanation.
9. Challenges and solutions.
10. Conclusion and future work.

C) Presentation (3 Marks)

1) **Project Overview**
   - Brief description of the project idea
   - Objectives and goals
2) **Graphics Concepts**
   - Explanation of the computer graphics concepts used
   - (transformations, lighting, textures, camera)
3) **System Demonstration**
   - Live demo of the project
   - Show key features and interactions
4) **Technical Implementation**
   - Overview of how the project was implemented
   - Tools, technologies, and programming approach
   - Highlight technical complexity
5) **Challenges and Solutions**
   - Main challenges faced during development
   - How they were solved
6) **Team Contribution**
   - Equal participation of all team members
7) **Presentation Quality**
   - Clear and well-designed slides
   - Good time management
8) **Discussion & Questions**
   - Ability to answer questions
   - Demonstration of technical understanding